

Applications of Flow and Circulations

- 1 Disjoint Paths in Graphs
 - computing different roads in traffic
 - the path decomposition procedure
- 2 The Survey Design Problem
 - consumer preference patterns
 - an algorithm for the survey design problem
- 3 Airline Scheduling
 - assigning airplanes to flights
 - the airline scheduling problem
 - scheduling with a feasible circulation in a flow network

CS 401/MCS 401 Lecture 17
Computer Algorithms I
Jan Verschelde, 27 July 2026

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disjoint paths

Consider the flow in a graph as traffic, where

- the edges represent roads, and
- the vertices are intersections between the roads.

Definition (1)

Two paths P and Q in a graph G are *edge disjoint* if P and Q share no edge.

Observe that the two paths may join vertices.

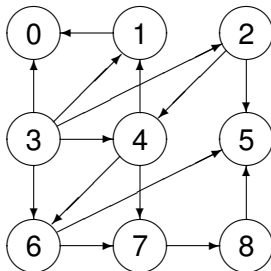
Can we travel from source to sink using different roads?

The Edge-Disjoint Paths problem asks to find the maximum number of edge-disjoint paths in a graph.

an example

$$V = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$$

$$E = \{(1, 0), (3, 0), (3, 1), (4, 1), (3, 2), (2, 4), (3, 4), (2, 5), (6, 5), (8, 5), (3, 6), (4, 6), (4, 7), (6, 7), (7, 8)\}$$

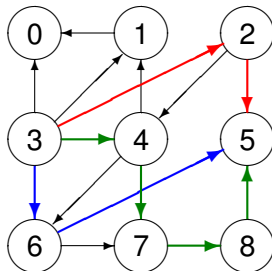


On how many edge disjoint paths can we travel from 3 to 5?

an example of three edge disjoint paths

$$V = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$$

$$E = \{(1, 0), (3, 0), (3, 1), (4, 1), (3, 2), (2, 4), (3, 4), (2, 5), (6, 5), (8, 5), (3, 6), (4, 6), (4, 7), (6, 7), (7, 8)\}$$



We can travel from 3 to 5 on three edge disjoint paths.

designing the algorithm

Let s and t be two vertices in a directed graph $G = (V, E)$.

We want to compute all edge-disjoint paths from s to t .

The flow network G' is defined as follows:

- 1 s is the source and t is the sink in G' .
- 2 set 1 as the capacities, for all $e \in E$: $c_e = 1$.

Lemma (2)

If there are k edge-disjoint paths in G from s to t , then the maximum flow in G' is at least k .

from edge-disjoint paths to flow

Lemma (2)

If there are k edge-disjoint paths in G from s to t , then the maximum flow in G' is at least k .

Proof. If we have k edge-disjoint paths in G , from s to t , then we define flow f as

- $f(e) = 1$ for all $e \in P$, for all paths P ,
- $f(e) = 0$ for all $e \in E$, e is not in any path.

This flow satisfies the capacity and conservation conditions.

The value of the flow is k , the number of paths.

Q.E.D.

The 0-1 valued flow can be interpreted as counting paths.

from flow to edge-disjoint paths

Lemma (3)

If f is a 0-1 valued flow of value F , then the set of edges with flow value $f(e) = 1$ contains a set of F edge-disjoint paths.

Lemma (2) and Lemma (3) imply the following.

Theorem (4)

There are k edge-disjoint paths in a directed graph G from s to t if and only if the value of the maximum flow of G is at least k .

Therefore, the Ford-Fulkerson algorithm applies.

The proof of Lemma (3) provides the *path decomposition procedure*.

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computing paths from flow

Lemma (3)

If f is a 0-1 valued flow of value F , then the set of edges with flow value $f(e) = 1$ contains a set of F edge-disjoint paths.

Proof. By induction on the number of edges that carry flow.

Base case: if no edges carry flow, then the number of edge-disjoint paths is zero. If $F = 0$, then no path from s to t has flow.

In the induction we compute a path P from s to t which carries flow.

- 1 We define a new flow f' : $f'(e) = 0$ for all $e \in P$.
- 2 The new flow f' has fewer edges and lower value $F - 1$.
- 3 Apply the induction hypothesis:
for value $F - 1$, we have $F - 1$ edge-disjoint paths.
- 4 Adding P to those $F - 1$ edge-disjoint paths proves the lemma.

proof of the lemma continues

We compute a path P from s to t which carries flow.

- 1 As $F > 0$, there is some vertex u : $f((s, u)) = 1$.
Set $f'((s, u)) := 0$ and $P := \{(s, u)\}$.
- 2 If $u = t$, then we have our path P .
- 3 If $u \neq t$, then, by the conservation conditions, there must be flow out of u .

So, there is some vertex v : $f((u, v)) = 1$, $f'((u, v)) := 0$
 $P := P \cup \{(u, v)\}$.

Apply step 2, with $u = v$.

What if we would have a cycle in P and reach the same vertex v before we arrive at the sink t ?

what if we have a cycle in P ?

A cycle C in our path P could cause an infinite loop!

Set all edges on C to zero, or use f' instead of f , as below.

- 1 Initialize the new flow f' to f .
- 2 At the first visit of vertex v , $f'((u, v)) = 1$: $f'((u, v)) := 0$.
- 3 If we visit v again, it will no longer be from u , as we have no longer flow from u to v , but from a different vertex w .
- 4 The number of vertices is bounded by n .

This algorithm does a *path decomposition of the flow* and this path decomposition procedure finishes the proof.

Q.E.D.

on the cost of the path decomposition procedure

Exercise 1:

Given a flow network G of n vertices and m edges.

What is the upper bound on the number of steps of the path decomposition procedure in function of n and m ?

Justify your bound.

running time

The Ford-Fulkerson algorithm on a graph with m edges and n vertices runs in time $O(mC)$ with $C = \sum_{e \text{ out of } s} c_e$.

Theorem (5)

The Ford-Fulkerson algorithm finds the maximum set of edge-disjoint paths from any two vertices in a graph with n vertices and m edges in $O(mn)$ time.

The value for C is the bound on the value of the maximal flow, which is for this problem bounded by n , the number of vertices.

We can solve the edge-disjoint paths problem in strongly polynomial time.

Menger's Theorem

The Ford-Fulkerson algorithm computes the maximum flow and the minimal cut.

Applying the max-flow min-cut theorem proves Menger's theorem.

Theorem (6)

In every directed graph with nodes s and t , the maximum number of edge-disjoint paths from s to t equals the minimum number of edges whose removal separates s from t .

Historically, similar to Hall's theorem we stated earlier, Menger's theorem was proven before the Ford-Fulkerson algorithm.

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consumer preference patterns

Given are

- q products,
- k consumers, and
- purchase histories for each consumer.

A **survey** asks consumers about a subset of q products.

Three guidelines to design the survey:

- 1 A consumer receives questions only about purchased products.
- 2 Consumer i is questioned about a number of products, a number between $\underline{c}_i$ and $\bar{c}_i$.
- 3 Product j receives replies from a distinct number of consumers, a number between $\underline{p}_j$ and $\bar{p}_j$.

the survey design problem

Given is a bipartite graph $G = (V, E)$,
 $V = A \cup B$, $A \cap B = \emptyset$:

- $A = \{ a_1, a_2, \dots, a_k \}$
is the set of k consumers,
- $B = \{ b_1, b_2, \dots, b_q \}$
is the set of q products.

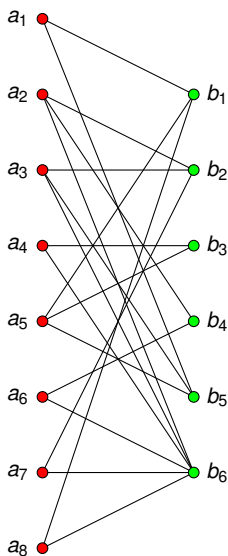
$E = \{ (a, b) \mid a \in A, b \in B \}$

For all $(a, b) \in E$:

consumer a purchased product b .

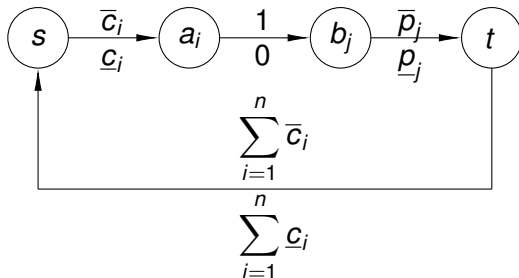
The survey design problem asks for

- $c_i \in [\underline{c}_i, \overline{c}_i]$, $\underline{c}_i < \overline{c}_i$,
 c_i product questions for customer i ,
- $p_j \in [\underline{p}_j, \overline{p}_j]$, $\underline{p}_j < \overline{p}_j$,
 p_j customer replies for product j .



a circulation problem on a flow network

Consider the flow network:



The flow on edge (a_i, b_j) is in $[0, 1]$:

- how many questions customer a_i is asked about product b_j .

The flow on edge (s, a_i) is in $[\underline{c}_i, \overline{c}_i]$:

- how many questions about products customer a_i receives.

The flow on edge (b_j, t) is in $[\underline{p}_j, \overline{p}_j]$:

- how many replies from consumers product b_j receives.

the survey design problem – an example

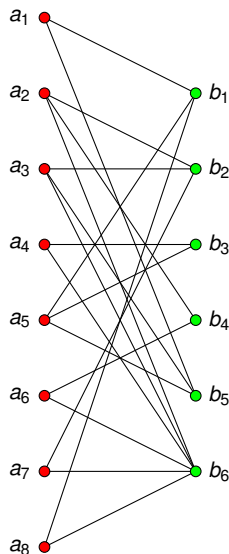
In the bipartite graph:

- $a_1, a_2, \dots, a_8$ are the consumers,
- $b_1, b_2, \dots, b_6$ are the products.

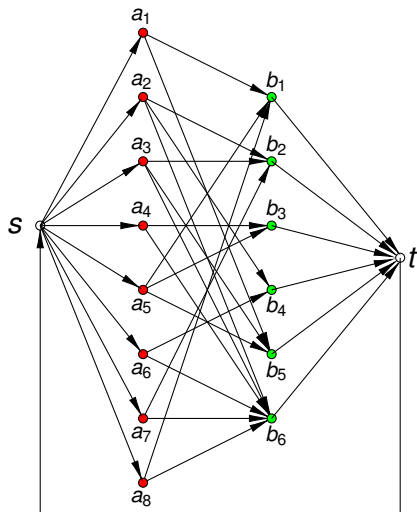
If a_i purchased b_j , then (a_i, b_j) is an edge.

Design the following survey:

- In the survey, every consumer is asked at least one question and no more than three questions.
- For each product, we want at least two replies of consumers and no more than three questions.



the flow network



Capacities on edges (a_i, b_j) :

- lower bounds are 0,
- upper bounds are 1.

Capacities on edges (s, a_i) :

- lower bounds are 1,
- upper bounds are 3.

Capacities on edges (b_j, t) :

- lower bounds are 2,
- upper bounds are 3.

Capacity on edges (t, s) :

- lower bound is 8,
- upper bound is 24.

a numerical exercise

Exercise 2:

We want to design a survey given the purchase histories of five consumers a , b , c , d , who purchased from seven different products, labeled 0 to 6, as encoded in the following tuples:

$(a, 0)$, $(a, 1)$, $(a, 2)$, $(b, 0)$, $(b, 1)$, $(b, 5)$, $(c, 0)$, $(c, 1)$, $(c, 4)$, $(c, 5)$,
 $(d, 0)$, $(d, 1)$, $(d, 2)$, $(d, 4)$, $(d, 6)$, $(e, 1)$, $(e, 4)$, $(e, 5)$.

- 1 In the survey we want to ask every consumer at least one question and no more than three questions.
- 2 For each product we want a number of replies between 2 and 3.

Show the equivalence with a flow problem.

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the survey design problem

Given is a bipartite graph $G = (V, E)$, $V = A \cup B$, $A \cap B = \emptyset$:

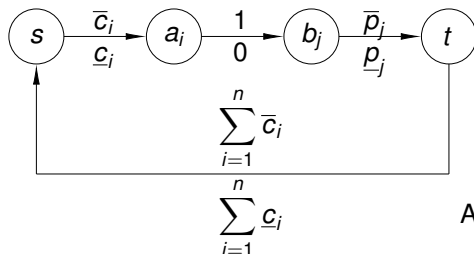
- $A = \{a_1, a_2, \dots, a_k\}$ is the set of k consumers,
- $B = \{b_1, b_2, \dots, b_q\}$ is the set of q products.

For all $(a, b) \in E$: consumer a purchased product b .

The survey design problem asks for

- $c_i \in [\underline{c}_i, \bar{c}_i]$, $\underline{c}_i < \bar{c}_i$, c_i product questions for customer i ,
- $p_j \in [\underline{p}_j, \bar{p}_j]$, $\underline{p}_j < \bar{p}_j$, p_j customer replies for product j .

With G corresponds the flow network G' :



All demands are zero.

the cost of constructing the flow network

Exercise 3:

Given the bipartite graph G for k consumers and q products,

- 1 define the algorithm to construct the flow network, and
- 2 justify the bound on the running time of the algorithm.

feasible circulations

Theorem (7)

The flow network G' has a feasible circulation if and only if there is a feasible way to design the survey.

Proof. If and only if means $\Rightarrow$ and $\Leftarrow$.

$\Rightarrow$: If there is a feasible circulation, then there is an integer valued solution for integer demands, capacities, and lower bounds.

An integer valued circulation naturally corresponds to a survey design: if the edge (a_i, b_j) carries a unit of flow, then consumer a_i will be surveyed about product b_j .

proof of the $\Leftarrow$ part

If we have a design for the survey, then we turn this design into a flow:

- If consumer a_i is asked about product b_j , then the edge (a_i, b_j) has one unit of flow, otherwise, there is no flow on the edge (a_i, b_j) .
- If consumer a_i is asked k questions, then the flow on (s, a_i) equals k .
- If k questions are asked about product b_j , then the flow on (b_j, t) equals k .
- The total number of questions is the flow on (t, s) .

end of the proof

The survey provides

- $c_i \in [\underline{c}_i, \bar{c}_i]$ product questions for consumer i , and
- $p_j \in [\underline{p}_j, \bar{p}_j]$ consumer replies for project j ;

which implies that all capacity conditions are satisfied on the flow.

If the survey satisfies the demand of zero at every vertex,
then the circulation also satisfies the demand of zero at all vertices.

Q.E.D.

the cost of the survey design problem

Exercise 4:

Given the bipartite graph G for k consumers and q products,

- 1 for $i = 1, 2, \dots, k$, given are two integers $\underline{c}_i, \bar{c}_i$, with $\underline{c}_i < \bar{c}_i$, lower and upper bounds on the questions for the consumers,
- 2 for $j = 1, 2, \dots, q$, given are two integers $\underline{p}_j, \bar{p}_j$, with $\underline{p}_j < \bar{p}_j$, lower and upper bounds on the questions on the products.

Express the cost of solving the survey design problem as a function of the input.

Can the problem be solved in strongly polynomial time?

an application: conference planning

Exercise 5:

A marketing director overseeing 5 representatives schedules their participation to 7 conferences, held at separate times in the year.

- 1 Every conference should be attended by at least one representative, and by at most three.
- 2 Every representative should attend at least two but not more than four conferences.

Under those constraints, is there a feasible schedule?
Justify your answer.

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assigning airplanes to flights

We have p airplanes and want to serve k flights.

Every flight has four parameters:

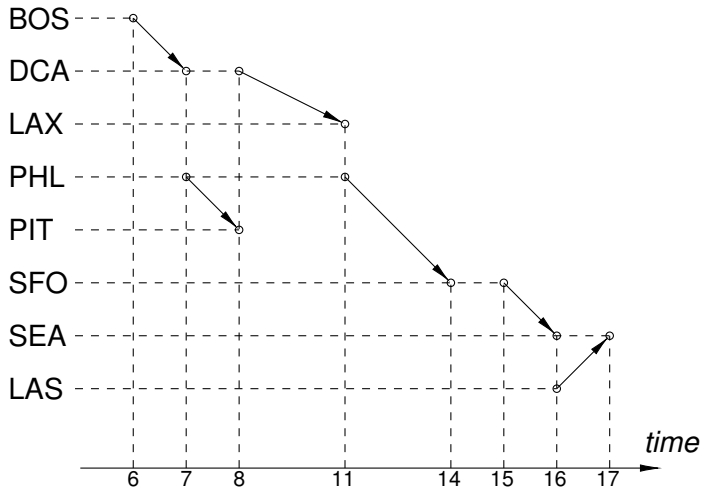
- 1 origin airport,
- 2 destination airport,
- 3 departure time, and
- 4 arrival time.

The same plane can serve flights i and j , but ...

- If the destination of i is the same as origin of j , then there should be enough time between the departure time of j and the arrival time of i .
- If the destination of i is *not* the same as origin of j , then the plane can fly from the destination of i to the origin of j , if there is enough time between the departure time of j and the arrival time of i .

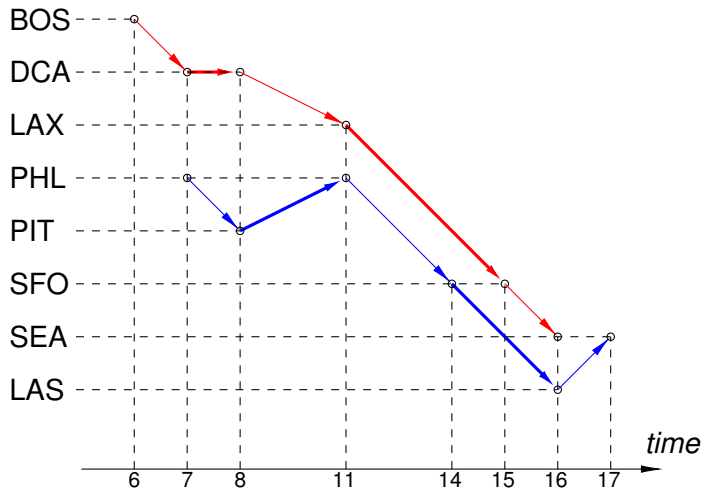
an example

Is it possible to serve 6 flights with 2 airplanes?

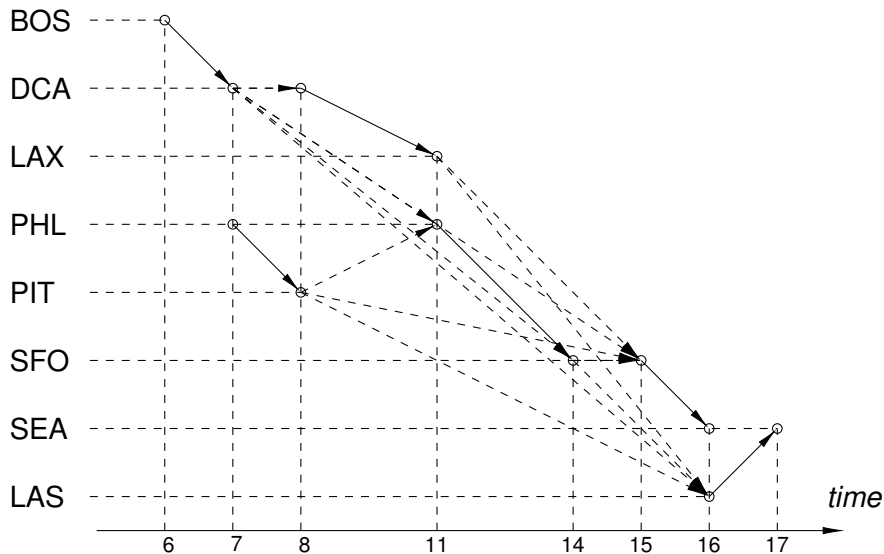


the example solved

Is it possible to serve 6 flights with 2 airplanes? Yes!



an expanded graph with reachable flights



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the airline scheduling problem

Definition (8)

Flight j is *reachable* from flight i if it is possible to use the same plane for flight i and then later for flight j .

We want to serve k flights $\{ 1, 2, \dots, k \}$, each specified by

- 1 origin airport,
- 2 destination airport,
- 3 departure time, and
- 4 arrival time.

In addition, the input contains the reachable pairs (i, j) .

The problem is to determine
if it is possible to serve all k flights with p planes?

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flights in the flow network

One unit of flow represents one airplane.

The flow network $G = (V, E)$ is defined as follows:

- Flight i is represented by
 - 1 two vertices u_i and v_i , its origin and destination,
 - 2 the edge $(u_i, v_i) \in E$,
 - 3 lower bound $\ell_{(u_i, v_i)} = 1$ and capacity $c_{(u_i, v_i)} = 1$.
- If flight j (represented by (u_j, v_j)) is reachable from flight i (represented by (u_i, v_i)), then $(v_i, u_j) \in E$ with $c_{(v_i, u_j)} = 1$ and $\ell_{(v_i, u_j)} = 0$.

source and sink in the flow network

The definition of the flow network G continues:

- The source s has demand $-p$, the sink t has demand p , where p is the number of airplanes.
All other vertices have demand 0.
- For each flight i , represented by (u_i, v_i) ,
 - 1 $(s, u_i) \in E$, with $c_{(s,u_i)} = 1$ and $\ell_{(s,u_i)} = 0$.
 - 2 $(v_i, t) \in E$, with $c_{(v_i,t)} = 1$ and $\ell_{(v_i,t)} = 0$.
- For extra planes: $(s, t) \in E$, $c_{(s,t)} = p$ and $\ell_{(s,t)} = 0$.

the flow network for the example

Exercise 6:

Consider the graph shown on slide 37.

- 1 Draw the flow network for this graph.
- 2 Consider the solution shown on slide 36.
 - 1 Show the solution on the flow network.
 - 2 Why is the solution optimal?
Use the termination criterion of the Ford-Fulkerson algorithm to justify your answer.

airline scheduling and feasible circulations

Theorem (9)

The k flights can be served by at most p planes if and only if there is a feasible circulation in the flow network G .

Proof. If and only if means $\Rightarrow$ and $\Leftarrow$.

$\Rightarrow$: If all flights can be served by $p' \leq p$ planes, then each plane defines a path P in G .

Through each path P , we send one unit of flow.

To satisfy all demands at s and t , send $p - p'$ units of flow on (s, t) .

The resulting circulation satisfies demand, capacity, and lower bound conditions.

proof of the $\Leftarrow$ part

Given a feasible circulation, the circulation is integer valued for integer capacities, lower bounds, and demands.

Assume k' units are on edges other than (s, t) .

As all other edges have a capacity of one, the circulation is integer, each such other edge carries exactly one unit of flow.

We convert a flow to a collection of paths as follows:

- Consider an edge (s, u_i) with one unit of flow.
- By conservation conditions, (u_i, v_i) carries one unit of flow.
- Continue the path P to t .

We repeat this construction and find p' paths from s to t .

Each path carries one unit of flow and represents the flight schedule for one airplane.

Q.E.D.

on the cost of airline scheduling

Exercise 7:

Given p , the number of planes, and given k flights, specified

- by origin and destination airports, and
- by departure and arrival times,

with the graph of reachable flights.

Express the cost of solving the airline scheduling problem as a function of the input.

Can this problem be solved in strongly polynomial time?

iterative algorithms

The Ford-Fulkerson algorithm solves the max flow problem in a weighted directed graph. It is an iterative algorithm.

As the cost of the algorithm depends on the size of the numbers, its running time is pseudo polynomial.

We applied this algorithm to five different problems:

- 1 computing the largest matching in a bipartite graph,
- 2 feasibility problem in circulation with demands,
- 3 edge-disjoint paths problem,
- 4 design of customer surveys, and
- 5 airline scheduling.

Many problems can be reduced to the max flow problem.